Chapter 3: Working with Financial Statements

Multiple Choice · Chapters 3, 5–7 · 55 questions · ★ = harder

1. Harborview Outfitters has market capitalization of $20,000 thousand (500 thousand shares at $40), interest-bearing debt of $1,760 thousand (notes payable plus long-term debt), cash of $260 thousand and EBITDA of $1,530 thousand. What is EV/EBITDA?

a. 13.07

b. 14.22

c. 14.05

d. 14.39

2. Firm X has a profit margin of 2%, total asset turnover of 4.0 and an equity multiplier of 2.5. Firm Y has a profit margin of 8%, total asset turnover of 1.0 and an equity multiplier of 2.5. Which statement is correct?

a. Firm X relies more on financial leverage than Firm Y.

b. Firm Y uses its assets more efficiently than Firm X.

c. Both earn an ROE of 20%: X through rapid asset turnover and Y through high margins, with equal leverage.

d. Firm Y earns the higher ROE because its margin is higher.

Chapter 3: Working with Financial Statements — Answers

1. c
Step 1: EV = Market cap + Interest-bearing debt − Cash
Step 2: EV = 20,000 + 1,760 − 260 = 21,500
Step 3: EV/EBITDA = 21,500 / 1,530 = 14.05
Why subtract cash: an acquirer also acquires the firm's cash.

2. c
Formula: ROE = Profit margin × Total asset turnover × Equity multiplier
Step 1: ROE(X) = 0.02 × 4.0 × 2.5 = 20%
Step 2: ROE(Y) = 0.08 × 1.0 × 2.5 = 20%
Step 3: Equity multipliers are equal (2.5), so leverage is the same
Reading: X turns assets over 4 times faster, Y earns a 4 times higher margin.

Chapter 5: Discounted Cash Flow Valuation

Multiple Choice · Chapters 3, 5–7 · 55 questions · ★ = harder

1. A lease requires $1,000 at the beginning of every month for 24 months. The rate is 6% compounded monthly. What is the lease worth today?

a. $22,562.87
b. $24,000.00
c. $22,675.68
d. $22,450.61

2. Lender A quotes 8.4% compounded monthly. Lender B quotes 8.5% compounded semiannually. Which loan is cheaper?

a. Lender A, because its quoted rate is lower
b. Lender A, because its EAR is 8.68% versus 8.73% for B
c. Lender B, because its EAR is 8.68% versus 8.73% for A
d. They cost the same

3. You save $400 at the end of every month in an account paying 6% compounded monthly. How many years until the balance reaches $30,000?

a. 63.85 years
b. 6.25 years
c. 5.32 years

4. You will receive $5,000 a year for 10 years, but the first payment arrives 4 years from today. The discount rate is 8%. What is the stream worth today?

a. $33,550.41
b. $24,660.55
c. $26,633.39
d. $28,764.07

5. You borrow $20,000 for 4 years with monthly payments. The bank quotes 8% compounded quarterly. What is the monthly payment?

a. $488.26
b. $416.67
c. $652.04
d. $487.76

Chapter 5: Discounted Cash Flow Valuation — Answers

1. c
Step 1: Monthly rate r = 6% / 12 = 0.5%
Step 2: PV (ordinary) = 1,000 × [1 − 1/(1.005)^{24}] / 0.005 = 22,562.87
Step 3: PV (due) = 22,562.87 × 1.005 = 22,675.68
BA II Plus: 24 N, 0.5 I/Y, −1000 PMT, 0 FV, CPT PV = 22,562.87
Then: × 1.005 = 22,675.68 (or BGN mode).

2. c
Step 1: EAR(A) = (1 + 0.084/12)^{12} − 1 = 8.73%
Step 2: EAR(B) = (1 + 0.085/2)^{2} − 1 = 8.68%
Step 3: Compare EARs, never quoted rates
Answer: B is cheaper (lower EAR).

3. c
Step 1: 30,000 = 400 × [(1.005)^{t} − 1] / 0.005
Step 2: (1.005)^{t} = 1.375
Step 3: t = ln 1.375 / ln 1.005 = 63.85 months
Step 4: Years = 63.85 / 12 = 5.32
BA II Plus: 0.5 I/Y, −400 PMT, 0 PV, 30000 FV, CPT N = 63.85 (months).

4. c
Step 1: The annuity formula gives the value one period before the first payment, at date 3
Step 2: PV_3 = 5,000 × [1 − 1/(1.08)^{10}] / 0.08 = 33,550.41
Step 3: PV_0 = 33,550.41 / (1.08)^{3} = 26,633.39
Watch for: discounting 4 years is the common error
BA II Plus: 10 N, 8 I/Y, −5000 PMT, 0 FV, CPT PV = 33,550.41
then 3 N, 0 PMT, −33550.41 FV, CPT PV = 26,633.39.

5. d
Step 1: Quarterly rate = 8% / 4 = 2%
Step 2: Effective monthly rate = (1.02)$^{1/3}$ − 1 = 0.6623%
Step 3: 20,000 = C × [1 − 1/(1.006623)^{48}] / 0.006623
Step 4: C = $487.76
Watch for: 8% / 12 assumes monthly compounding and gives $488.26
BA II Plus: 48 N, 0.6623 I/Y, 20000 PV, 0 FV, CPT PMT = −487.76.

Chapter 6: Interest Rates and Bond Valuation

Multiple Choice · Chapters 3, 5–7 · 55 questions · ★ = harder

1. A firm issued a bond at par with an 8% coupon. Market interest rates have since risen. Which statement is correct?

a. The coupon rate rises to match the market, so the price stays at par.
b. The coupon rate stays at 8%, the YTM rises, and the price falls below par.
c. The YTM stays at 8% because it is set at issue.
d. The face value falls to restore the 8% yield.

2. A $1,000 bond pays an 8% coupon semiannually and matures in 6 years. The yield to maturity is 10% (quoted annually). What is its price?

a. $591.18
b. $911.37
c. $912.89
d. $1,000.00

3. A $1,000 bond pays an 8% annual coupon, matures in 3 years and now sells at par. If its yield rises to 9%, by how much does its price change?

a. −1.00%
b. −2.53%
c. −3.00%
d. +2.62%

4. A bond's yield can either rise by 2 percentage points or fall by 2 percentage points. Which statement about the price change is correct?

a. The price changes by the same dollar amount in either direction.
b. The price falls more when yields rise than it rises when yields fall.
c. The price rises more when yields fall than it falls when yields rise.
d. The price does not change because the coupon is fixed.

5. Four bonds currently yield 8%. Which bond's price changes most when the market yield rises by one percentage point?

a. 5-year maturity, 10% coupon
b. 5-year maturity, 4% coupon
c. 20-year maturity, 10% coupon
d. 20-year maturity, 4% coupon

6. A bond's quoted yield to maturity equals the return an investor actually earns only under which conditions?

a. The bond is held to maturity, it does not default, and coupons are reinvested at the YTM.
b. The bond is sold before maturity at par.
c. Interest rates fall after purchase.
d. The coupon rate equals the current yield.

7. A $1,000 bond pays a 6% coupon semiannually, matures in 10 years and sells for $925. What is its yield to maturity (quoted annually)?

a. 3.53%
b. 6.49%
c. 7.06%
d. 7.18%

8. A bond with an 8% coupon rate and 8 years to maturity trades below par. Which ordering of its yields is correct?

a. YTM < current yield < coupon rate
b. Current yield < coupon rate < YTM
c. Coupon rate = current yield = YTM
d. Coupon rate < current yield < YTM

9. Compared with an otherwise identical non-callable bond, a callable bond should have which characteristics?

a. A higher price and a lower yield
b. A lower price and a higher yield
c. The same price and yield
d. A higher price and a higher yield

10. A municipal bond yields 4.2% and a comparable taxable bond yields 6.0%. Three investors face marginal tax rates of 20%, 30% and 35%. Which statement is correct?

a. All three prefer the taxable bond.

b. All three prefer the muni.

c. Only the 35% investor prefers the muni; the 30% investor is indifferent.

d. Only the 20% investor prefers the muni.

11. A $1,000 bond pays an 8% coupon semiannually. Two months have passed since the last coupon, and the bond is quoted at $985. What is the invoice (dirty) price?

a. $971.67

b. $985.00

c. $998.33

d. $1,025.00

12. A bond yields 11.3% and expected inflation is 6%. What is the exact real rate?

a. 5.30%

b. 5.00%

c. 17.98%

13. A $1,000 bond pays a 6% coupon semiannually and matures in 10 years. Investors require an effective annual return of 8.16%. What is the bond's price?

a. $1,000.00

b. $854.26

c. $864.10

d. $856.10

14. How does interest rate risk change with maturity?

a. It falls as maturity rises.

b. It rises with maturity at an increasing rate.

c. It rises with maturity, but at a decreasing rate.

d. It does not depend on maturity.

15. Among bonds with the same maturity, which is the most sensitive to interest rate changes?

a. A bond with a 15% coupon

b. A bond with a 10% coupon

c. A bond with a 5% coupon

d. A zero coupon bond

16. Expected inflation rises, but nothing changes about the issuer's credit or the real return investors require. What happens to the bond's price?

a. Nothing, because the coupon is fixed.

b. It rises, because the coupon is worth more.

c. It falls, because nominal yields rise with expected inflation.

d. It rises, because the real rate falls.

17. Compared with a Treasury bond of the same maturity, which premiums explain the higher yield on a corporate bond?

a. Real rate and inflation premium

b. Interest rate risk premium only

c. Default risk, taxability and liquidity premiums

18. Which pairing of bond type and feature is correct?

a. Put bond: lets the issuer buy the bond back, lowering its value

b. Convertible bond: the holder may exchange it for shares, so it carries a lower coupon

c. Floating-rate bond: the coupon is fixed for life

d. Income bond: coupons are paid even if the firm has no income

Chapter 6: Interest Rates and Bond Valuation — Answers

1. b
Coupon rate: set at issue and fixed for the life of the bond
YTM: set by the market and changes every day
Price: whatever makes the two consistent
Here: higher YTM with a fixed coupon means a lower price.

2. b
Step 1: Convert to periods: $t = 6 \times 2 = 12$
$C = 80 / 2 = 40$
$r = 10\% / 2 = 5\%$
Step 2: PV of coupons $= 40 \times [1 - 1/(1.05)^{12}] / 0.05 = 354.53$
Step 3: PV of face value $= 1{,}000 / (1.05)^{12} = 556.84$
Step 4: Price $= 354.53 + 556.84 = 911.37$
BA II Plus: 12 N, 5 I/Y, 40 PMT, 1000 FV, CPT PV = −911.37.

3. b
Step 1: At 8% the price is \$1,000 (coupon rate = YTM)
Step 2: At 9%: Price $= 80 \times [1 - 1/(1.09)^{3}] / 0.09 + 1{,}000 / (1.09)^{3} = 974.69$
Step 3: Change $= 974.69 / 1{,}000 - 1 = -2.53\%$
Rule: prices and yields move in opposite directions.

4. c
Shape: the price-yield curve is convex
Result: prices rise faster when yields fall than they drop when yields rise by the same amount
Note: interest rate risk is price risk, not default risk.

5. d
Maturity effect: a longer bond has more cash flows exposed, and the distant face value swings most
Coupon effect: a lower coupon pushes more value to maturity, so it is more sensitive
Answer: the 20-year, 4% coupon bond combines both effects.

6. a
YTM: the single discount rate that sets the PV of the promised cash flows equal to the price (the bond's IRR)
Assumption 1: held to maturity
Assumption 2: no default (YTM is a promised yield)
Assumption 3: coupons reinvested at the YTM
If rates fall, reinvested coupons earn less and the realized return falls short.

7. c
Step 1: $t = 20$ periods
$C = 30$
Price $= 925$
Step 2: Solve $925 = 30 \times [1 - 1/(1 + r)^{20}] / r + 1{,}000 / (1 + r)^{20}$
Step 3: $r = 3.53\%$ per six months
Step 4: Quoted YTM $= 3.53\% \times 2 = 7.06\%$
BA II Plus: 20 N, −925 PV, 30 PMT, 1000 FV, CPT I/Y = 3.53
Check: price below par, so YTM > coupon rate.

8. d
Coupon rate = Annual coupon / Face value
Current yield = Annual coupon / Price
it is higher because the price is below face
YTM: also counts the capital gain, so it is highest
Discount bond: coupon rate < current yield < YTM.

9. b
Step 1: The issuer calls when rates have fallen
Step 2: Holders lose the high coupon exactly when they would most like to keep it
Step 3: So investors demand a higher yield
Result: a higher yield means a lower price.

10. c
Formula: Equivalent taxable yield $= 4.2\% / (1 - t)$
20%: $4.2 / 0.80 = 5.25\% < 6.0\%$, taxable is better
30%: $4.2 / 0.70 = 6.00\%$, indifferent
35%: $4.2 / 0.65 = 6.46\% > 6.0\%$, muni is better
Lesson: munis are held mainly by high-bracket investors.

11. c
Step 1: Coupon per period $= 80 / 2 = 40$
Step 2: Accrued interest $= 40 \times (2 / 6) = 13.33$
Step 3: Dirty price = Clean price + Accrued interest $= 985.00 + 13.33 = 998.33$
Why: the buyer reimburses the seller for the part of the next coupon the seller earned.

12. b
Formula: $1 + R = (1 + r) \times (1 + h)$
Step 1: $1 + r = 1.113 / 1.06 = 1.05$
Step 2: $r = 5.00\%$
Note: the approximation $R - h = 5.30\%$ is not exact, and the error grows with inflation.

13. c
Step 1: Semiannual rate $= (1.0816)^{1/2} - 1 = 4.00\%$
Step 2: $t = 20$
$C = 30$
Step 3: Price $= 30 \times [1 - 1/(1.04)^{20}] / 0.04 + 1{,}000 / (1.04)^{20} = 864.10$
Watch for: $8.16\% / 2 = 4.08\%$ treats an effective rate as a quoted rate and gives \$854.26
BA II Plus: 20 N, 4 I/Y, 30 PMT, 1000 FV, CPT PV = −864.10.

14. c
Rule: longer bonds are hit harder by rate changes
But: the effect rises at a decreasing rate
Example: going from 1 to 10 years matters far more than going from 20 to 30 years.

15. d
Rule: for a given maturity, the lower the coupon, the greater the price sensitivity
Reason: a low-coupon bond delivers most of its value at maturity, far in the future
Extreme case: a zero coupon bond has all of its value at maturity.

16. c
Bond yields are nominal
Fisher effect: higher expected inflation raises nominal yields
Higher yield with fixed promised cash flows means a lower price.

17. c
Term structure: real rate, inflation premium and interest rate risk premium are common to both bonds
Bond-specific: default risk, taxability and liquidity premiums
A corporate bond carries these and the Treasury largely does not.

18. b
Convertible: exchangeable for a fixed number of shares at the holder's option
the right is valuable, so the coupon is lower
Put bond: the holder may force the issuer to buy it back
it raises the bond's value
Floater: coupon resets against a reference rate, often with a floor and ceiling (collar)
Income bond: coupons paid only if income is sufficient.

Chapter 7: Equity Markets and Stock Valuation

Multiple Choice · Chapters 3, 5–7 · 55 questions · ★ = harder

1. A company just paid a dividend of $2.00. Dividends grow at 5% a year forever and the required return is 12%. What will the stock be worth five years from now?
   a. $36.47
   b. $30.00
   c. $38.29
   d. $40.20

2. The required return on a stock is 9%. The next dividend is $1.00 and the growth rate is assumed to be 10% forever. What does the dividend growth model say?
   a. The price is $100, so the stock is undervalued.
   b. The price is $11.11 because growth is ignored when g exceeds R.
   c. The price is −$100, which shows the assumption g > R cannot hold forever.

3. A firm just paid a dividend of $2.00. Dividends grow at 15% a year for the next 3 years and at 5% a year forever after that. The required return is 12%. What is the stock worth today?
   a. $30.00
   b. $35.32
   c. $38.80
   d. $45.63

4. A firm will pay no dividend for the next 3 years. It will pay $1.20 at the end of year 4, and that dividend will then grow at 5% forever. The required return is 13%. What is the stock worth today?

a. $15.00
b. $9.20
c. $10.40
d. $0.74

5. Which pairing of growth case and its watch-out is correct?

a. Zero growth: $D_1 = D_0$ holds in every case.
b. Constant growth: use D_0 in the numerator.
c. Nonconstant growth: discount P_t by t + 1 periods.
d. Constant growth: use D_1 and require R > g.

6. Which statement about preferred stock is correct?

a. Failing to pay a preferred dividend triggers default and bankruptcy.
b. Preferred dividends are tax-deductible to the issuer.
c. Preferred holders rank ahead of all creditors in liquidation.
d. Preferred dividends are not tax-deductible, and skipping one is not a default.

7. A firm just paid a dividend of $2.00. Dividends grow at 8% a year for 5 years and at 4% a year forever after that. The required return is 10%. What is the stock worth today?

a. $34.67
b. $9.47
c. $50.94
d. $41.10

8. A stock sells for $50. It just paid a dividend of $2.00, and investors require 11%. What constant growth rate is the market implying?

a. 4.00%
b. 7.00%
c. 6.73%
d. 5.50%

9. A stock sells for $36. It just paid a dividend of $1.80, and dividends grow at 5% forever. What is the required return?

a. 10.00%
b. 10.25%
c. 5.25%
d. 15.00%

10. A firm earns an ROE of 15% and pays out 40% of its earnings as dividends. Next year's earnings per share are expected to be $3.00, and investors require 13%. What is the stock worth?

a. $12.86
b. $20.00
c. $30.00
d. $13.33

11. The next dividend is $2.00 and dividends grow at 5% forever. If the required return rises from 10% to 11%, by how much does the stock price change?

a. −1.0%
b. −9.1%
c. −16.7%
d. −50.0%

12. A stock sells for $60. Investors require 12% and dividends grow at 8% forever. What dividend did the firm just pay?

a. $2.40
b. $4.80
c. $2.22
d. $7.20

13. A firm pays out 50% of its earnings. Dividends grow at 6% forever and the required return is 12%. What price-earnings ratio (price ÷ next year's earnings) does the dividend growth model imply?

a. 6.00
b. 16.67
c. 4.17
d. 8.33

14. Han River Utilities has preferred stock with a $100 par value and an annual dividend rate of 7.5%, paid quarterly. The dividend never changes. Investors require 9%, compounded monthly. What is the price of one preferred share?

a. $83.33
b. $82.71
c. $79.95
d. $250.00

15. Sejong Holdings has just issued preferred stock that pays a fixed $4.50 per year forever, but the first dividend will not be paid until three years from today. The required return is 12%. What is the price today?

a. $26.69
b. $33.48
c. $37.50
d. $29.89

16. Daehan Foods just paid a dividend of $1.80. Dividends grow at 6% forever and investors require 13%. What are (a) the price today and (b) the price four years from today?

a. $25.71; $32.46
b. $27.26; $34.41
c. $27.26; $32.46
d. $27.26; $36.48

17. Mirae Logistics paid a dividend of $1.20 four years ago, and the dividend just paid was $1.62. Dividends will keep growing at the same average annual rate. Investors require 14%. What is the price today?

a. $33.56
b. $26.09
c. $28.13

18. Bitgoeul Biotech just paid a dividend of $2.00. Dividends grow 25% next year, 18% the year after and 12% in year 3, then 6% forever. The required return is 14%. What is the price today?

a. $32.61
b. $34.57
c. $26.50
d. $36.24

19. Hanbit Robotics will pay no dividend for the next three years. It will pay $2.00 at the end of year 4 and $2.50 at the end of year 5, after which dividends grow at 4% forever. The required return is 15%. What is the price today?

a. $14.14
b. $21.50
c. $13.69
d. $12.29

20. Gyeongbok Software just paid a dividend of $1.50. Dividends grow at 15% a year for five years and at 5% forever after that. The required return is 16%. Using the two-stage model, what is the price today?

a. $20.37
b. $21.02
c. $14.32
d. $19.13

21. Seorabeol Pharma just paid a dividend of $3.00. Dividends grow at 10% a year for eight years and then at 3% forever. The required return is 11%. What is the price today?

a. $57.93
b. $38.63
c. $23.05
d. $58.97

22. Namsan Electronics just reported EPS of $3.20, and EPS is expected to grow 8% next year. It pays no dividend. Comparable firms trade at forward P/E ratios of 14, 17, 13 and 16. Using the average benchmark, what should the stock sell for?

a. $48.00
b. $51.84
c. $58.75
d. $55.30

23. Cheongdam Mobility had sales of $480 million and a net loss of $36 million. It has 24 million shares and pays no dividend. Comparable firms have price-sales ratios of 2.1, 2.6, 1.9 and 2.4, and an average P/E of 22. Estimate the stock price.

a. −$33.00
b. $48.00
c. $45.00
d. $1,080.00

24. A firm just paid a dividend of $1.00. Dividends grow at 30% for the next two years, at 15% in year 3, and at 5% forever after that. The required return is 13%. What is the price today?

a. $20.66
b. $21.50
c. $19.47
d. $25.51

25. A firm just reported EPS of $2.50, and EPS is expected to grow 10% next year. Comparable firms trade at trailing P/E ratios (price divided by the most recent EPS) of 18, 20, 16 and 22. Using the average benchmark, what should the stock sell for?

a. $52.25
b. $47.50
c. $45.00
d. $55.00

26. Why is a share of stock harder to value than a bond?

a. Stock valuation uses a different principle than bond valuation.
b. Dividends are contractual obligations, but the discount rate is unknown.
c. The market price of a stock can be used to back out its required return exactly.
d. Dividends are discretionary, there is no maturity date, and both R and g must be estimated.

27. A firm pays no dividend now. It is expected never to pay any dividend, and no one who would pay one will ever buy it. What does the dividend discount model imply?

a. The shares are valued at book value.
b. The model cannot be used, so only P/E applies.
c. The shares are worthless.
d. The shares are worth the discounted sales.

28. An investor says: "I am not buying this stock for its dividends, I am buying it to sell it higher." Which response is correct?

a. The dividend discount model does not apply to such investors.

b. The model values only stocks held to maturity.

c. The model already contains this: the future price is itself the present value of later dividends.

d. Capital gains are not part of the model.

29. A board skips the common dividend for several years. What is the legal consequence?

a. The firm is in default.

b. Creditors may demand immediate repayment.

c. The unpaid dividends become a debt of the firm.

d. None: dividends are discretionary, so skipping them cannot cause default or bankruptcy.

30. Which statement distinguishes a dealer from a broker?

a. A broker holds inventory and a dealer earns only a commission.

b. A dealer buys and sells for its own account and bears the risk; a broker matches trades for a commission and takes no position.

c. Both take the other side of the customer's trade.

d. A dealer arranges trades and never holds shares.

Chapter 7: Equity Markets and Stock Valuation — Answers

1. c
Rule: the model works from any date, using the dividend that follows
Step 1: $D_6 = 2.00 \times (1.05)^6 = 2.68$
Step 2: $P_5 = D_6 / (R - g) = 2.68 / 0.07 = \$38.29$
Check: $P_0 = 2.10 / 0.07 = 30.00$ and $P_0 \times (1.05)^5 = 38.29$.

2. c
Step 1: $P_0 = D_1 / (R - g) = 1.00 / (0.09 - 0.10) = -\$100$
Step 2: A negative or infinite price means the assumption is impossible
Rule: the model requires $R > g$
no firm can outgrow its required return forever.

3. c
Step 1: $D_1 = 2.30$
$D_2 = 2.645$
$D_3 = 3.04175$
Step 2: $D_4 = 3.04175 \times 1.05 = 3.1938$
Step 3: $P_3 = D_4 / (R - g) = 3.1938 / 0.07 = 45.63$
Step 4: PV of dividends = 2.0536 + 2.1086 + 2.1651 = 6.33
Step 5: PV of $P_3 = 45.63 / (1.12)^3 = 32.48$
Step 6: $P_0 = 6.33 + 32.48 = \$38.80$.

4. c
Step 1: Apply the model one period before the first dividend, at date 3
Step 2: $P_3 = D_4 / (R - g) = 1.20 / (0.13 - 0.05) = \$15.00$
Step 3: PV $= 15.00 / (1.13)^3 = \$10.40$
Watch for: the model gives the price at date 3, so discount 3 periods, not 4.

5. d
Zero growth: $D_1 = D_0$ here, and only here
Constant growth: use D_1 in the numerator and require $R > g$
Nonconstant growth: discount P_t by t, not t + 1
Multiples: everything depends on the word "comparable".

6. d
Legal form: preferred stock is equity
Tax: the dividend is paid from after-tax income and is not deductible
Default: unpaid dividends are not a debt, so skipping one cannot cause default
Ranking: preferred holders rank behind every creditor.

7. d
Step 1: $D_1 = 2.00 \times 1.08 = 2.16$
Stage 1 PV $= D_1 / (R - g_1) \times [1 - ((1 + g_1) / (1 + R))^5]$
Stage 1 PV $= 2.16 / 0.02 \times [1 - (1.08 / 1.10)^5] = 9.47$
Step 2: $D_6 = 2.00 \times (1.08)^5 \times 1.04 = 3.06$
Step 3: $P_5 = D_6 / (R - g_2) = 3.06 / 0.06 = 50.94$
Step 4: PV of $P_5 = 50.94 / (1.10)^5 = 31.63$
Step 5: $P_0 = 9.47 + 31.63 = \$41.10$.

8. c
Setup: $P_0 = D_0 \times (1 + g) / (R - g)$
Step 1: $50 = 2.00 \times (1 + g) / (0.11 - g)$
Step 2: $5.50 - 50g = 2.00 + 2.00g$
Step 3: $52g = 3.50$, so $g = 6.73\%$
Check: $D_1 = 2.1346$ and $2.1346 / (0.11 - 0.0673) = 50.00$.

9. b
Formula: $R = D_1 / P_0 + g$
Step 1: $D_1 = 1.80 \times 1.05 = 1.89$
Step 2: Dividend yield $= 1.89 / 36 = 5.25\%$
Step 3: $R = 5.25\% + 5.00\% = 10.25\%$
Watch for: the dividend just paid is D_0, so grow it first.

10. c
Step 1: Retention ratio $b = 1 - 0.40 = 0.60$
Step 2: $g = ROE \times b = 0.15 \times 0.60 = 9\%$
Step 3: $D_1 = 0.40 \times 3.00 = 1.20$
Step 4: $P_0 = D_1 / (R - g) = 1.20 / (0.13 - 0.09) = \$30.00$
Idea: growth comes from earnings the firm retains and reinvests at its ROE.

11. c
Step 1: At R = 10%: $P_0 = 2.00 / (0.10 - 0.05) = \$40.00$
Step 2: At R = 11%: $P_0 = 2.00 / (0.11 - 0.05) = \$33.33$
Step 3: Change $= 33.33 / 40.00 - 1 = -16.7\%$
Lesson: the price is highly sensitive to R and g because R − g is a small number.

12. c
Step 1: Dividend yield $= R - g = 12\% - 8\% = 4\%$
Step 2: $D_1 = 0.04 \times 60 = \$2.40$
Step 3: $D_0 = D_1 / (1 + g) = 2.40 / 1.08 = \$2.22$
Watch for: \$2.40 is next year's dividend, not the one just paid.

13. d
Step 1: $P_0 = D_1 / (R - g)$ and $D_1 = \text{Payout} \times E_1$
Step 2: $P_0 / E_1 = \text{Payout} / (R - g)$
Step 3: $P/E = 0.50 / (0.12 - 0.06) = 8.33$
Reading: higher growth or a lower required return raises the justified P/E.

14. b
Step 1: Quarterly dividend $D = 0.075 \times 100 / 4 = \$1.875$
Step 2: Quarterly rate $R = (1 + 0.09/12)^3 - 1 = 2.2669\%$
Step 3: $P_0 = D / R = 1.875 / 0.022669 = \$82.71$
Watch for: $0.09 / 4 = 2.25\%$ ignores monthly compounding and gives \$83.33.

15. d
Step 1: $P = D / R$ gives the value one period before the first payment, at date 2
Step 2: $P_2 = 4.50 / 0.12 = \$37.50$
Step 3: $P_0 = 37.50 / (1.12)^2 = \$29.89$
Watch for: discounting \$37.50 by three years gives \$26.69.

16. b
Step 1: $P_0 = D_0(1 + g) / (R - g) = 1.80 \times 1.06 / 0.07 = \$27.26$
Step 2: $P_4 = D_5 / (R - g) = 1.80 \times (1.06)^5 / 0.07 = \$34.41$
Check: $27.26 \times (1.06)^4 = 34.41$, because the price grows at g.

17. c
Step 1: $1.20 \times (1 + g)^4 = 1.62$, so $g = (1.35)^{1/4} - 1 = 7.79\%$
Step 2: $D_1 = 1.62 \times 1.0779 = 1.7462$
Step 3: $P_0 = 1.7462 / (0.14 - 0.0779) = \$28.13$
Watch for: 35% / 4 = 8.75% is a simple average and overstates g.

18. d
Step 1: $D_1 = 2.50$
$D_2 = 2.95$
$D_3 = 3.304$
Step 2: $P_3 = D_4 / (R - g) = 3.304 \times 1.06 / 0.08 = 43.778$
Step 3: $P_0 = 2.50 / 1.14 + 2.95 / (1.14)^2 + (3.304 + 43.778) / (1.14)^3 = \$36.24$
Watch for: P_3 uses D_4 and is discounted 3 years, not 4.

19. a
Step 1: $P_5 = D_6 / (R - g) = 2.50 \times 1.04 / 0.11 = 23.636$
Step 2: $P_0 = 2.00 / (1.15)^4 + (2.50 + 23.636) / (1.15)^5 = \$14.14$
Watch for: the three zero years must be counted (CF: F01 = 3), or every later cash flow is discounted too few periods.

20. b
Step 1: First stage = $[1.50 \times 1.15 / (0.16 - 0.15)] \times [1 - (1.15 / 1.16)^5] = 7.308$
Step 2: $D_5 = 1.50 \times (1.15)^5 = 3.0170$
Step 3: $P_5 = 3.0170 \times 1.05 / 0.11 = 28.799$
Step 4: $P_0 = 7.308 + 28.799 / (1.16)^5 = 7.308 + 13.712 = \$21.02$
Watch for: $R - g_1 = 1\%$ is tiny, so keep full precision.

21. d
Step 1: First stage = $[3.00 \times 1.10 / (0.11 - 0.10)] \times [1 - (1.10 / 1.11)^8] = 23.047$
Step 2: $D_8 = 3.00 \times (1.10)^8 = 6.4308$
Step 3: $P_8 = D_9 / (R - g_2) = 6.4308 \times 1.03 / 0.08 = 82.796$
Step 4: $P_0 = 23.047 + 82.796 / (1.11)^8 = 23.047 + 35.927 = \$58.97$
Watch for: P_8 is built from D_9, not D_8.

22. b
Step 1: Benchmark P/E = (14 + 17 + 13 + 16) / 4 = 15
Step 2: $EPS_1 = 3.20 \times 1.08 = 3.456$
Step 3: $P_0 = 15 \times 3.456 = \$51.84$
Watch for: a forward P/E must be applied to next year's EPS
$15 \times 3.20 = \$48.00$ mismatches them.

23. c
Step 1: Sales per share = 480 / 24 = \$20.00
Step 2: Benchmark P/S = (2.1 + 2.6 + 1.9 + 2.4) / 4 = 2.25
Step 3: $P_0 = 2.25 \times 20.00 = \$45.00$
Why not P/E: EPS = −36 / 24 = −\$1.50, so $22 \times (-1.50)$ gives a meaningless negative price.

24. b
Step 1: $D_1 = 1.30$
$D_2 = 1.69$
$D_3 = 1.9435$
Step 2: $P_3 = D_4 / (R - g) = 1.9435 \times 1.05 / 0.08 = 25.508$
Step 3: $P_0 = 1.30 / 1.13 + 1.69 / (1.13)^2 + (1.9435 + 25.508) / (1.13)^3 = \$21.50$
Watch for: P_3 uses D_4 and is discounted 3 years, not 4.

25. b
Step 1: Benchmark P/E = (18 + 20 + 16 + 22) / 4 = 19
Step 2: A trailing P/E is applied to the most recent EPS: \$2.50
Step 3: $P_0 = 19 \times 2.50 = \$47.50$
Watch for: applying a trailing P/E to next year's EPS ($19 \times 2.75 = \$52.25$) mismatches the ratio and the earnings.

26. d
Same principle: value = PV of expected cash flows
Problem 1: a dividend is declared at the board's discretion and can be cut to zero without default
Problem 2: a share has no maturity, so the stream is infinite with no face value to anchor it
Problem 3: both the required return and the growth rate must be estimated, and the price is very sensitive to both.

27. c
Principle: price = PV of every dividend the share will ever pay
Firm that eventually pays: the dividend may be far off, but it must be there
Firm that never pays and is never bought by anyone who would: no cash flow to discount
Answer: the shares are worthless.

28. c
Step 1: One-period holding value $P_0 = (D_1 + P_1) / (1 + R)$
Step 2: $P_1 = (D_2 + P_2) / (1 + R)$
Step 3: Substitute again and again
the price terms disappear
Result: only dividends remain
The objection is contained within the model, not a refutation of it.

29. d
Dividends are declared at the board's discretion
Unpaid dividends are not a debt of the firm
Result: failing to pay them cannot cause default or bankruptcy
Contrast: a missed coupon is a default.

30. b
Dealer: trades for its own account, holds an inventory, takes the other side of your trade and bears the risk
Broker: arranges a trade between two other parties, never takes a position
Pay: a broker earns a commission for matching, not for bearing risk.